



EXAMINATION PAPER

FACULTY : COMPUTER SCIENCE AND MULTIMEDIA
COURSE : BACHELOR OF INFORMATION TECHNOLOGY (HONS)
YEAR/ SEMESTER : SECOND YEAR / SEMESTER FOUR
MODULE TITLE : COMPLIER DESIGN AND CONSTRUCTION
CODE : BIT 354
TIME ALLOWED : 3 HOURS

Instruction to candidates

1. This question paper has THREE (3) Sections.
2. Answer **ALL** questions in Section A, VSAQ.
3. Answer **5** questions in Section B, SAQ.
4. Answer **3** questions in Section C, LAQ.
5. No scripts or answer sheets are to be taken out of the Examination Hall.

Do not open this question paper until instructed

(Candidates are required to give their answers in their own words as far as practicable)

SECTION A
Very Short Answer Questions
Attempt All Questions

[10×2=20]

1. Describe compiler construction briefly.
2. Present the role of a lexical.
3. Define Syntax Directed Definition (SDD).
4. What is basic block in code optimization?
5. Differentiate between token and lexeme.
6. List the different phases of a compiler.
7. Illustrate concept of type checking in compiler design.
8. Define a symbol table and its purpose.
9. Write the concept of back patching.
10. What are terminals and non-terminals in CFG?

SECTION B
Short Answer Questions
Attempt any five (5) questions out of six (6) questions [5×7=35]

1. Explain your understanding about Syntax Analysis. Calculate the first and follow functions for the given grammar: [2+5]
 $S \rightarrow ABC$
 $A \rightarrow aB / Ad$
 $B \rightarrow b/g/h$
 $C \rightarrow c$
2. Discuss in detail the main design issues in a code generator and how they affect code quality.
3. Examine the importance of error handler in the compiler. How is it manipulated in the different phases of compilation? [3+4]

4. Define syntax in the context of programming languages. Explain syntax-directed translation (SDT) and how it is used in a compiler. Illustrate your answer with a suitable example. [2+5]
5. State about intermediate code and three address codes. Write a three-address code for the expression $a = -b*(c+d)$ [4+3]
6. Explain the role of a lexical analyzer in detail. Also describe input buffering technique. [4+3]

SECTION C

Long Answer Questions

Attempt any three (3) questions out of four (4) questions. [3×15=45]

1. Compare Top-Down and Bottom-Up parsing techniques. Construct LL (1) parsing table of following grammar. And check Input: $id + id * id$ [5+10]
$$E \rightarrow T E'$$
$$E' \rightarrow + T E' \mid \epsilon$$
$$T \rightarrow F T'$$
$$T' \rightarrow * F T' \mid \epsilon$$
$$F \rightarrow (E) \mid id$$
2. Define context-free grammar and demonstrate writing grammar for simple arithmetic expressions. Give a detail definition of NFA and DFA with example and convert the regular expression $(a + \epsilon) b^*$ into an equivalent DFA using a direct method. [3+12]
3. Explain the different phases of a compiler with a neat diagram. Briefly describe the role of each phase and give an example to illustrate how source code progresses through the compiler. [7+8]

4. Differentiate between synthesized and inherited attributes. Give a detail definition on L-attributed and S-attributed. Write Syntax Directed Definition to carry out type checking for the following expression. $[4+4+7]$
- $E \rightarrow id \mid E1 \text{ op } E2 \mid E1 \text{ relop } E2 \mid E1[E2] \mid E1 \uparrow$
- (where op is an arithmetic operator and relop is a relational operator).

******BEST OF LUCK******